The Impact of Climate & Artificial Change on Agriculture

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December 2024

1 Introduction

Climate change is a phenomenon that occurs on multiple time scales, often affecting human activities; while part of climate change is dominated by physically processes such as weather, El Nino, the North Atlantic Oscillation, Asian Monsoon variations, droughts, and circulations of the atmosphere and oceans, a significant part of climate change is also due to anthropogenic change, resulting in global warming, ozone holes, deforestation, pollution and other ecosystem changes. Thus, as a result of climate change, rising temperatures, shifting precipitation patterns, and frequent weather events are threatening the stability and productivity of agricultural systems worldwide, leading to decreased crop yields and food security and disruptions in water availability.

Furthermore, research from the Environmental Protection Agency and Oxfam emphasizes how the unpredictability of climate conditions are making it harder for farmers, especially in developing countries where agriculture is a primary source of food, to manage crop production effectively and therefore exacerbating global hunger due to heightened risks of crop failures from intense climate change phenomenon. A study by NASA further warns of significant declining yields in major crops such as corn and wheat, highlighting the consequences for global food supply chains and the urgency for farmers and policymakers to strategize and adapt to different agricultural practices as a result of climate change. Moreover, The Economist adds how climate change is expected to shift agricultural zones – for example, crops such as coffee and grape face reduced yields in traditional growing areas while new regions such as northern Europe could be more prone to harvest crops such as maize. Consequently, the article points to global trade as a measure to accommodate for the geographic shift in agriculture due to various climate change phenomena.

Through data visualizations such as a geographical analysis, clustering, and predictive models, we can illustrate the impacts of climate change on agriculture, allowing analysis on complex patterns and trends across various geographic regions across the globe. Through further visual and exploratory analysis, we

can highlight the relationships between different climate change variables and agricultural outcomes.

2 Project Goals

By analyzing this dataset, we seek to answer:

How are different crops and regions affected by various climatic factors – specifically, what shifts in climate or artificial factors such as fertilizer, pesticide use, etc. are likely to impact food security and farming practices across the globe in the future? How do we see this relationship differ not only from a global perspective, but also from country to country? What insightful interactions between climate or artificial variables and agricultural outcomes can we deduce to help provide not only valuable information to farmers, but also for policymakers working to strategize against climate change to mitigate impacts on food production? What common combinations of climate and artificial variables from our clustering and predictive models help notify shortages in advance, enabling for actions such as importing food, etc. in a timely manner?

3 Data Description

The dataset used in this project provides a comprehensive view of the worldwide agricultural system and its relationship with environmental and artificial factors. Especially, we focus on exploring the influence on crop yields across various countries and regions. The raw dataset contains many key columns: Year: The time span of observations from 1990 to 2024. Country: 10 countries with various climates and agriculture practices include India, China, France, Canada, USA, Argentina, Australia, Nigeria, Russia, and Brazil. Crop type: 10 major crop types including corn, wheat, coffee, sugarcane, fruits, rice, barley, vegetable, soybeans, and cotton. Environmental features: Average-temperature-C: Annual average temperature Total-precipitation-mm: Annual total precipitation CO2-Emissions-MT: Carbon emissions Extreme-Weather-Events: Count of severe weather events Soil-Health-index: Natural soil quality Artificial features: Irrigation-Access: Percentage of farmland with irrigation systems Pesticide-Use-KG-per-HA: Pesticide used per hectare Fertilizer-Use-KG-per-HA: Fertilizer used per hectare Adaptation-Strategies: Measures taken by humans to adapt to climate change impacts Outcome feature: Crop-yield-MT-per-HA: Crop Yield metric tons per hectare Economic-Impact-Million-USD: Economic consequences of agricultural shifts

3.1 Data Pre-Processing

We first checked the dataset for missing values. Then, we handled outliers using the Interquartile Range(IQR) method. The raw data itself seems very tidy with no null values and outliers. Also, we notified the dataset includes a time-related

column named Year. To make time-based analysis easier and more efficient, the year was converted into a data type, and rearranged the dataset accordingly. Additionally, we identified Economic-Impact-Million-USD as a dependent variable. Since it was not directly relevant to the intended analysis for our project, this column was removed to streamline the dataset. After that, we utilized Min-Max scaling to normalize features including 'Average-Temperature-C', 'Total-Precipitation-mm', 'CO2-Emissions-MT', 'Irrigation-Access-', 'Pesticide-Use-KG-per-HA', 'Fertilizer-Use-KG-per-HA', 'Soil-Health-Index', 'Crop-Yield-MT-per-HA' to bring them to the same scale.

3.2 Initial Exploration of Dataset and Features

We checked the distribution of data count and average crop yield across crop types to ensure whether any crop type is underrepresented.

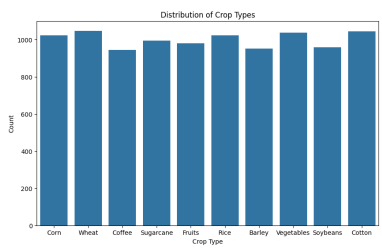


Chart 1: Crop Distribution

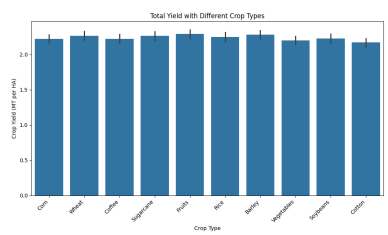


Chart 2: Average Crop Yield

The above bar charts display the distribution of crop and average crop yield across different crop types. The consistent yield across crop types emphasizes the importance of exploring other environmental variables, and human practices that may affect agricultural productivity. Additionally, the even distribution of crop types ensures any relationships identified during the analysis are not skewed due to overrepresentation or underrepresentation of specific crop types. This strengthens the validity of the conclusions that we have drawn from this dataset.

To analyze local factors influencing agricultural trends, we plotted the crop types with the most significant fluctuations over time, as well as the total yield across different countries throughout the same period.
According to Charts 3 and 4, each country exhibits a distinct pattern in yield variability. For example, Australia and Brazil show high yield fluctuations, possibly due to variable climate conditions. India and China have demonstrated more stable yields and are significant agricultural producers, likely due to diverse cropping systems that balance overall yield. This observation suggests that our next focus should be analyzing crop yield variability across different regions.

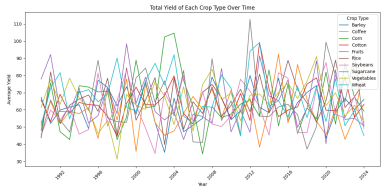


Chart 3: Total Yield of Crop Type Over Time

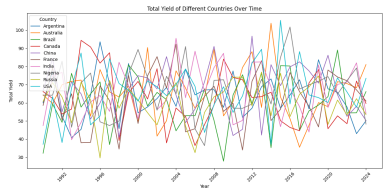


Chart 4: Total Yield of Different Country Over Time

For the early exploratory analysis, we applied the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) to examine how crop yields are influenced by the previous year's yield. Our analysis revealed no clear cyclical patterns, except for dependence on lag 1 in the case of rice yields. To better understand how different variables are interrelated, we analyzed the correlation matrix to examine the strength and direction of relationships between the features. The analysis aimed to identify pairs of variables with strong correlations that might influence crop yields. From the correlation matrix, we observed there is no significant correlation pattern between individual environmental or human-induced features and crop yields. For example, the soil health index, fertilizer use, average temperature, and irrigation didn't have a noticeable correlation with crop yields. We have explored pairwise relationships between features, but the findings indicate that the single variables may not strongly predict crop yields. It's likely a combination of factors(both environmental and artificial) influenced agricultural productivity.

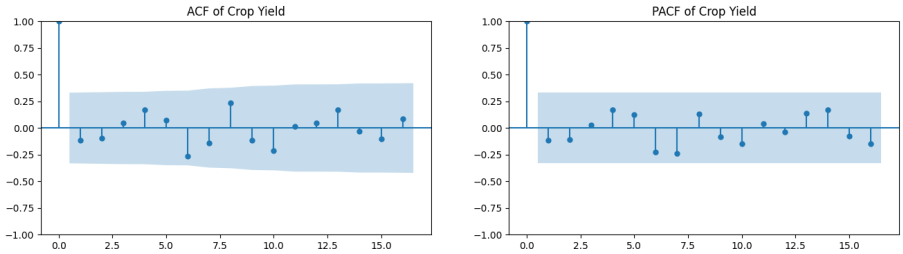


Chart 5: ACF/PACF of global average yield, alpha =0.05

– As we can see in the ACF/PACF plot, no lags seems to exceed the critical value, this analysis look similar on specific countries/crops as well, as exmained in our notebook

Due to the lack of significant pairwise correlations, we propose conducting clustering analysis to identify which clusters are most impactful and to understand why certain clusters exhibit better or worse yields. Additionally, we plan to apply nonlinear models to analyze key factors influencing crop yields and make yield predictions. Furthermore, we aim to perform a geographical analysis to

achieve a comprehensive understanding of crop yield variability across different regions.

4 Exploratory Analysis: Geographical Approach

We begin our exploratory analysis by performing a geographical analysis, on both a country and regional level, to help pinpoint what climate or artificial features contribute the most to crop yields on a microscopic level. Later on in our clustering and predictive models, we will visualize patterns between the two on a global, macro level and see if our results align.

4.0.1 Basic Feature Analysis by Country

To start this analysis, we first perform a basic feature analysis, where for each crop, we utilize seaborn to plot the relationship between the average climate (for example, average temperature, precipitation, etc.) or artificial variable and crop yield over the given time period in the dataset. As mentioned, the function, basic feature analysis, produces these plots for each crop, where each plot highlights the relationship between the two variables for each country. For the sake of being concise and consistent in this report, we choose to analyze patterns for the crop rice but note that we can generalize our analysis for the rest of the crops (as seen by the visualizations in our code). Additionally, in the figures below, we pick to visualize the trends between average temperature, average precipitation, average, extreme events, and average irrigation access VS. the crop yield for a more concise analysis, but emphasize how the results found can be generalized to all other climate/ artificial features (as seen in our code) except adaptation strategies.

In Figure 1, we see that for average temperature, there is significant variability in how different countries respond; however, we note that the optimal yield is around moderate temperatures, about 14 to 16 degrees celsius. We can attribute this behavior to concluding how heat tolerance for rice might differ across the globe and can be influenced by other regional climate conditions, farming practices, and artificial features such as irrigation or fertilizer use. We see a similar but more defined trend in Figure 2 for precipitation, where once again, we have a variability in different countries, where some have a positive, inconsistent, or even negative correlation between the two. This is surprising as one would think that, in general, a greater precipitation would lead to greater crop yields. Thus, this suggests how other features might have a stronger effect on the average crop yield per country. We observe similar patterns for both average extreme events and average irrigation access in Figures 3 and 4, where the relationship between the two variables is quite noisy. In all cases, we can conclude that the relationships between the rich yields and climate or artificial variables are country specific, point to a non-linear relationship between these variables and crop yields. This result will later help us build appropriate models later on in this section.

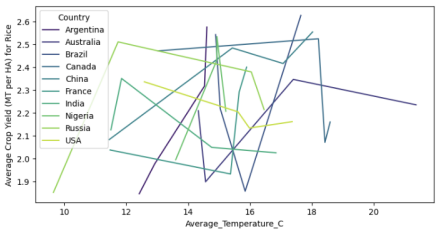


Chart 6: Average Temperature

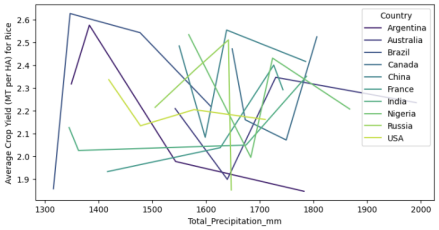


Chart 7: Average Precipitation

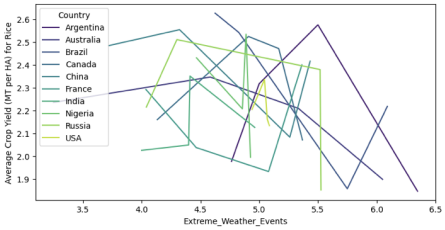


Chart 8: Average Extreme Events

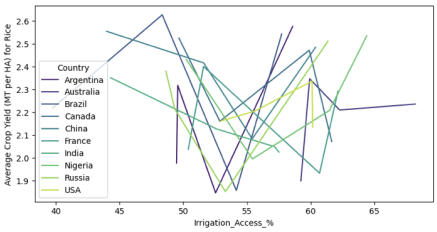


Chart 9: Average Irrigation Access

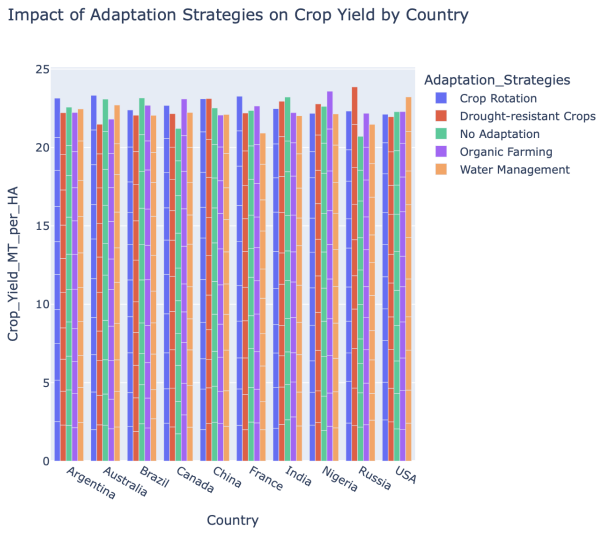


Chart 10: Adaptation Strategies for each Country and Crop

As we mentioned earlier, we do not generalize this pattern to adaptation strategies as by the bar plot in Figure 5, each country seems to generate a similar crop yield for each crop for a given strategy. We can thus conclude that adaptation strategies should not necessarily have the greatest impact on crop yields for a specific crop (across different countries) when seeking out a combination of factors that greatly affect yields. We will check this after building the rest of our models.

4.0.2 Visualizations of Crop Yields on a Regional Basis

To better understand the non-linear combination of climate or artificial features that affects crop yields in each country, we perform a regional analysis to observe the concentration of crop yields for a particular crop in different regions of a country. For consistency, we choose rice; in our code, we visualize these regional visualization for each crop. Moreover, for these visualizations, we merge the dataset with their corresponding geoBoundaries geojson file found online. We are only successful in visualizing the heatmaps for India, Canada, and Brazil as the information for the two datatsets for the other countries does not match as needed; however, to successfully analyze how trends in the regional analysis for our chosen crop, rice, affects the non-linear combination of features by country, we will perform a feature importance in the next section to deepen our understanding of which features significantly affect rice yields in different parts of the globe.

In Figure 6, we see that Northwestern and Western India have the highest rice yields. Due to India's tropical temperatures, monsoon seasons, and vast size, it is expected that not all regions have similar crop yields for rice. In our code, we retrieve this visualization for every crop in India and note similar behaviors for all crops: most have a specific region in the country where they flourish. On the other hand, in Figure 7, we see that Eastern Canada, in Quebec, has the greatest rice yields. Given that eastern Canada has brutal cold winters and is on the other side of the world map, it is expected that a different combination of factors (environmental and artificial) affect rice yields in the two countries. In the next section, we perform an in depth feature importance by region for each crop and country; from this regional analysis, we will deduce for a specific crop, which combination of features leads to the highest crop yield and lowest crop yield per country. We do this by summarizing the feature importance regional analysis for each country. For consistency in this report, we choose rice, but through our code we can make a similar conclusion for other crops.

4.0.3 Feature Importance by Region

As stated above, we perform our feature analysis by crop for each region in a specified country by training a Random Forest Regressor model. In our code, we display the results in the form of a bar plot. For consistency, we choose rice as our crop of choice and analze the regional behavior across the different

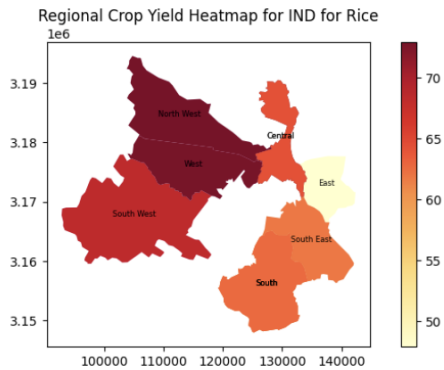


Chart 11: India

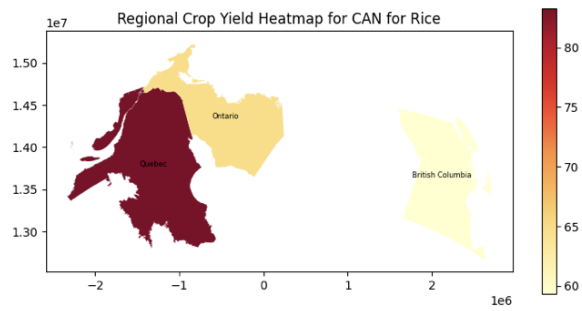


Chart 12: Canada

countries in our datatset. While all visualizations (for all countries) are present in our code, in this report we only include a few plots for example.

Starting with the eastern side of the globe, we see that in India, we have four main regions: Eastern (West Bengal), Southern (Tamil Nadu), Northern Western (Punjab), and Western (Maharashtra). In the earlier section, we saw how North Western and Western India had the greatest rice yields - from our visualization in Figure 8, we can attribute this behavior due to, significantly, CO2 emissions as well as total precipitation and average temperature. In Eastern and Southern India, we see similar patterns where CO2 emissions and total precipitation contribute the most to rice yields whereas in Northern India, only total precipitation seems to be a strong contender. In China, we see that in the North, soil moisture and fertilizer use are the key features, with similar results in the South, but instead of fertilizer use, average temperature plays a bigger role (North & South have the greatest rice yields). Average temperature, alongside CO2 emissions, are the main features in Eastern China while in the central region, we see total precipitation having the greatest impact on rice yields.

Moving towards the northern part of the globe, in Russia, in regions with abundant crop yields, the Siberian and Central areas of the country, we consistently see the importance of CO2 emissions, total precipitation, as well as total extreme events. In other regions, we see similar patterns in addition to importance of soil health. On the other hand, in France, specifically in areas with high rice yields, we see that total precipitation, average temperature, and irrigation access have the most significant impact on rice yields in the country.

In the South, we have Brazil, Argentina, Brazil, Nigeria, and Australia: in regions which produce the largest yields, we see a common combination of average temperature, fertilizer use, and soil health in Australia and Nigeria as well as CO2 emissions and average temperature or CO2 and fertilizer/pesticide use in Brazil and Argentina respectively.

Lastly, in the Northern hemisphere, we have USA and Canada: in the US, we see a combination of irrigation access, fertilizer use, and average temperature/CO2 emissions producing the greatest yields while in Canada we see a similar pattern of a combination of average temperature, CO2 emissions, and soil health index.

For the most part, on a macro level, we see common trends between a combination of average temperature, total precipitation, and CO2 emissions having the greatest importance on rice yields, with soil health being a strong feature throughout the globe. We note, on average, artificial features such as fertilizer/pesticide use and irrigation access, while used across the various countries, has a stronger influence on yields in the Western hemisphere, while climate change variables such as temperature, rain, and carbon dioxide emissions have a global impact. In the next section, we use clustering and predictive models to provide further evidence to these macro trends.

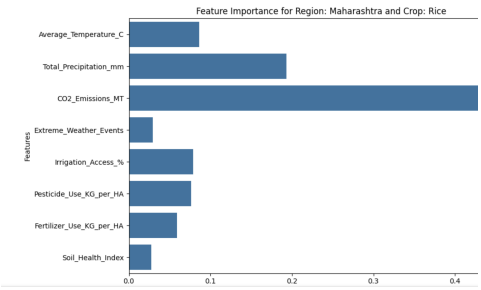


Chart 13: Western India

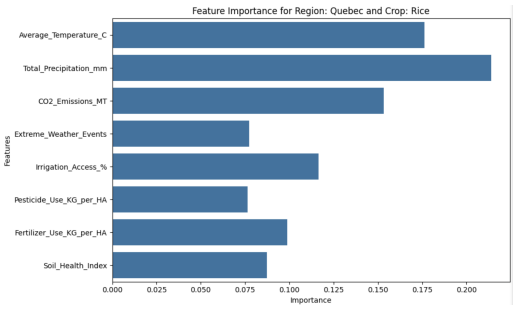


Chart 14: Eastern Canada

5 Clustering Analysis

We want to further explore patterns in the dataset like what combinations of features would be more likely to happen on instances. Also from the **correlation matrix** we have seen that the features has few correlations to yield, so we want to make clusters as our feature construction to predict. Hence we employed clustering techniques on both natural and artificial features. The clustering methods used were:

- **K-Means Clustering**: A linear method that partitions the data into a fixed number of clusters by minimizing the variance within each cluster.
- **Spectral Clustering**: A non-linear method leveraging graph-based techniques for clustering complex data structures.

We adopted these two models because they use number of clustering as hyperparameters, and we want to control the number of clusters we pick to 3 - 5 (Considering each clustering (natural and artificial) takes in 4 features) and

avoid one - encoding clustering leads to **overly sparse** data. For each model, we manually try different hyperparameters to optimize our average yield difference and data seperation through visualization of our PCA reduced data.

5.1 Data Standardization

To ensure fair comparison and effective clustering, all numeric columns were standardized using the `StandardScaler` from `sklearn`.

$$z = \frac{x - \mu}{\sigma}$$

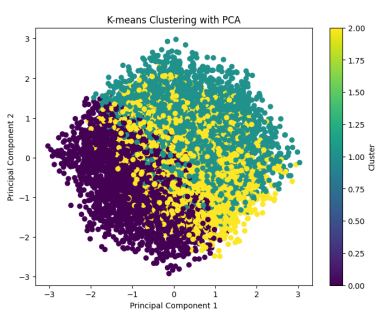
where μ is the mean and σ is the standard deviation of the characteristic.

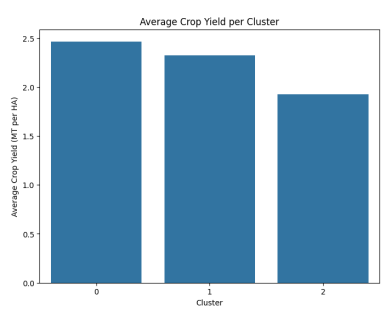
5.2 Clustering on Natural Features

We selected the following natural features for clustering:

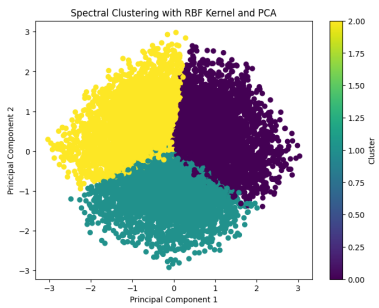
- Average Temperature ($^\circ C$)
- Total Precipitation (mm)
- Extreme Weather Events
- Soil Health Index

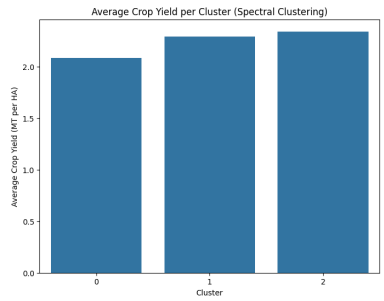
K-Means Clustering Results Using 3 clusters, we found seperations and meaningful difference in average yield, as shown below:





Spectral Clustering Results With an RBF kernel and 3 clusters, spectral clustering also provided cluster-wise separations which looks even more seperated, but smaller difference in yield:



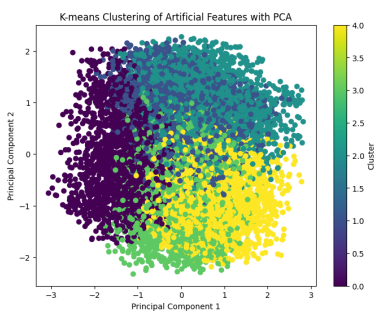


Clustering on Artificial Features

We selected the following artificial features:

- CO_2 Emissions (MT)
- Pesticide Use (KG/HA)
- Fertilizer Use (KG/HA)
- Irrigation Access (%)

K-Means Clustering Results Using 5 clusters, this method provided the most distinctive separations and difference in average yield. As we can see, the result is not as ideal as that of natural clusters because the clusters seperates worse and has less difference in yields. The results are shown below:



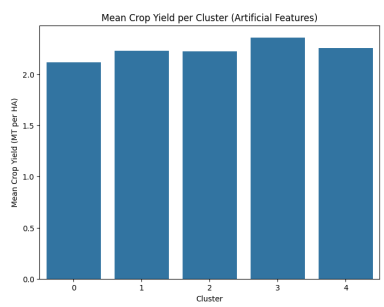
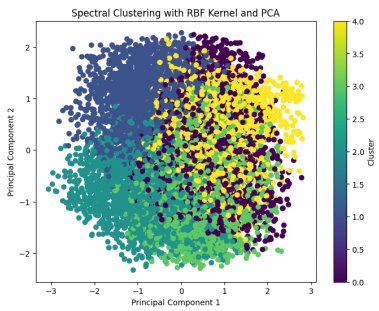
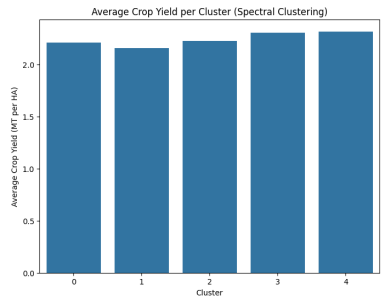


Chart 15: Average Irrigation Access

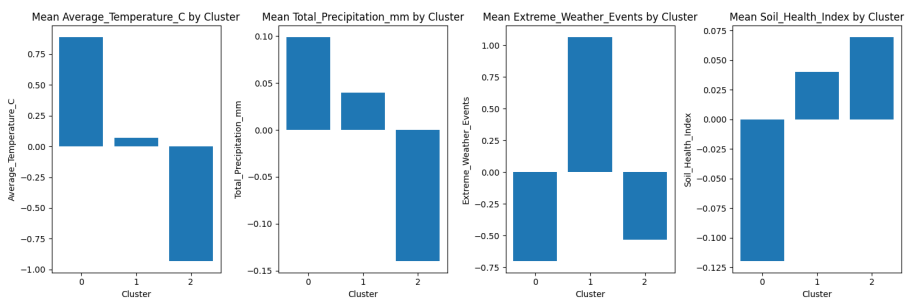
Spectral Clustering Results Using an RBF kernel and 5 clusters, spectral clustering results are depicted below, again the results are not ideal, and it generates even less difference than kmeans in artificial features.

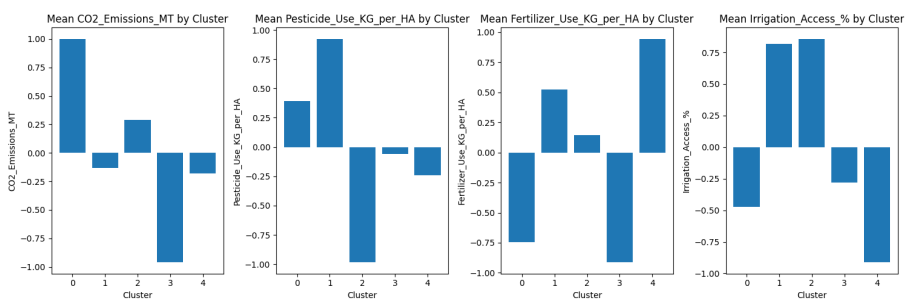




5.3 Cluster Insights

Previous bar plots were used to visualize average crop yield for each cluster. **Kmeans** seems to outperform **spectral** in both natural and artificial, so we decided to use kmeans clusters as our **features for prediction**.
Besides prediction, we are also curious about each cluster's feature values, indicating what combinations of conditions are more common.





Interpretation

Natural Conditions Clustering The clustering based on natural conditions (average temperature, total precipitation, extreme weather events, and soil health index) reveals clear distinctions among the clusters. Cluster 1 is associated with higher average temperatures, moderate precipitation, and better soil health. Cluster 2 stands out with the highest levels of extreme weather events

but lower soil health, suggesting regions prone to climatic instability. Cluster 0, on the other hand, shows poor soil health and lower precipitation, which could indicate areas facing challenges in natural fertility and water availability.
Artificial Conditions Clustering The clustering of artificial conditions (CO 2 2 emissions, pesticide use, fertilizer use, and irrigation access) highlights varying levels of human influence on agricultural practices. Cluster 4 shows the highest CO 2 2 emissions and fertilizer use, indicating regions with intensive industrial or agricultural activities. Cluster 1 exhibits the highest pesticide usage, pointing to areas of high agricultural dependency. Conversely, Cluster 0 has the lowest irrigation access, suggesting potential water management challenges, while other clusters demonstrate varying degrees of human-induced impact on farming systems.

6 Prediction

From our exploration, we found that the variance of yield is very high of 1 unit out of an average value of 2.2 unit. Thus, a shock of lower yield may lead to shortage of food. If we can use our features to predict the yield, we can predict low yield to take actions like restoring or importing food in advance.

1. Preprocessing

To prepare the dataset for modeling, we applied the following preprocessing steps:

- **Categorical Variables**: Standardizations Already done in clustering
- **Categorical Variables**:
  - **Mean Target Encoding**: Applied to `Crop_Type` and `Country` based on their average crop yields, since they have more than 10 types, doing one-hot encoding will make the training data too sparse.
  - **One-Hot Encoding**: Applied to `Adaptation Strategies` and clustering features from K-means and Spectral Clustering,features having less than 5 types.
- **Train-Test Split**: The data was split into an 80%-20% ratio for training and testing

2. Linear Regression

Linear Regression was implemented as a baseline model due to its simplicity and interpretability. The Mean Squared Error (MSE) for LR was calculated as:

$$MSE = 0.89$$

which is slightly lower than our target's variance of around 1

Prediction Fit The following plot compares the actual crop yields with the predictions made by the LR model, we can see the fit is pretty bad:

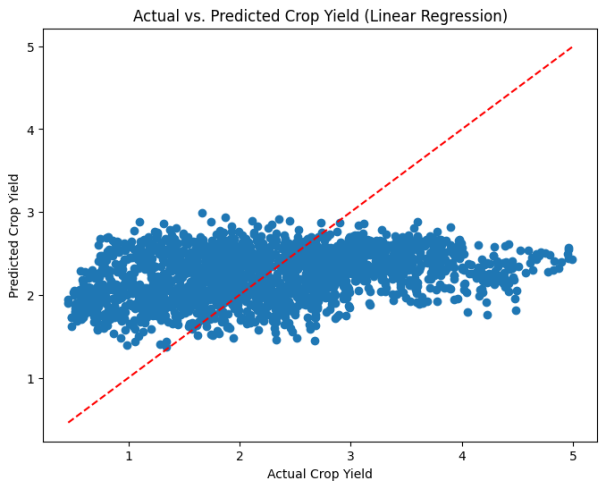


Chart 16: Actual vs. Predicted Crop Yield (Linear Regression)

Feature Importance The feature importance in Linear Regression is derived from the model's coefficients, we can see temperature and exterme weather seems most significant:

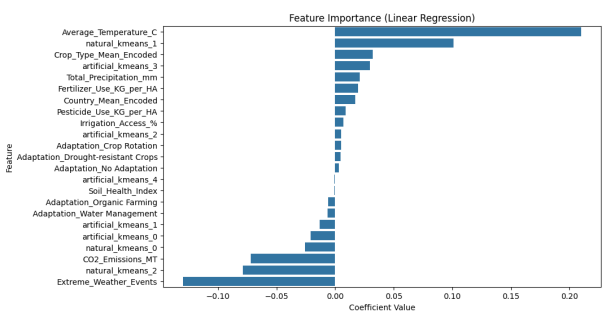


Chart 17: Feature Importance in Linear Regression (Coefficient Values)

3. XGBoost (XGB)

Since LR seems to underfit our data, we used XGBoost, a powerful ensemble-based algorithm, was implemented with hyperparameters optimized using Bayesian Optimization (Since it has a wide range of hyperparameters). The optimized model achieved a significantly lower MSE of:

$$MSE = 0.68$$

This is much lower than variance of 1, but remains much unpredictable variances

Prediction Fit The prediction fit is shown below, although some variance remains unexplained (which is fair because all the feature have low correlations to yield), it demonstrates better alignment between actual and predicted values compared to LR:

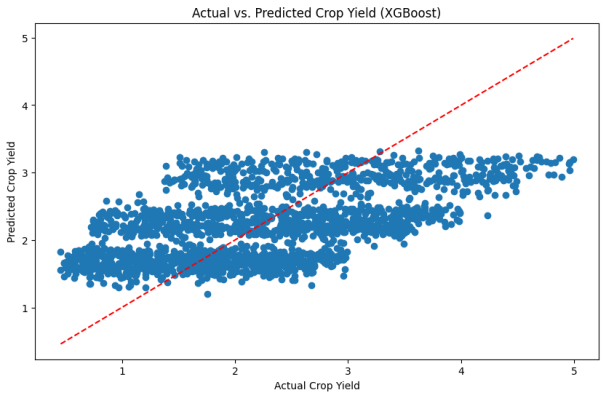


Chart 18: Actual vs. Predicted Crop Yield (XGBoost)

Feature Importance XGBoost provides a natural measure of feature importance through its tree structure. As we can see below, tempertature is extermely important. personally, I guess that XGBoost finalizes trees that use **temperature** as the first split factor, and itsimportance lead to the 3-platforms structure in our prediction plot

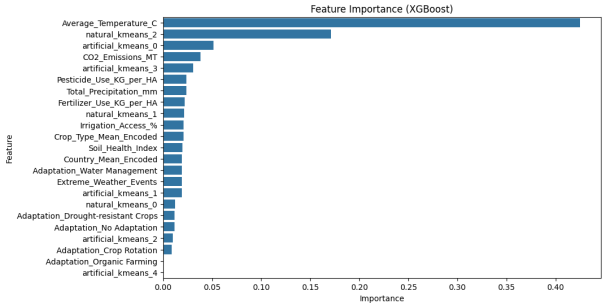


Chart 19: Feature Importance in XGBoost

4. Fully Connected Neural Network (FNN)

We the try to use models with more parameters like neural networks. Through our analysis there is little effect of adding beyond 3 layers or 128 nodes. An FNN model was implemented with three hidden layers of decreasing sizes (128, 64, and 16 neurons) with Relu activation. Initially, we were facing severe overfitting as seen the train/validation loss over epochs below:

Chart 20: Training and Validation Loss (FNN)

$$MSE = 0.8766$$

Hence we applied dropout layers and retrained the model, which seems to overcome overfitting well:

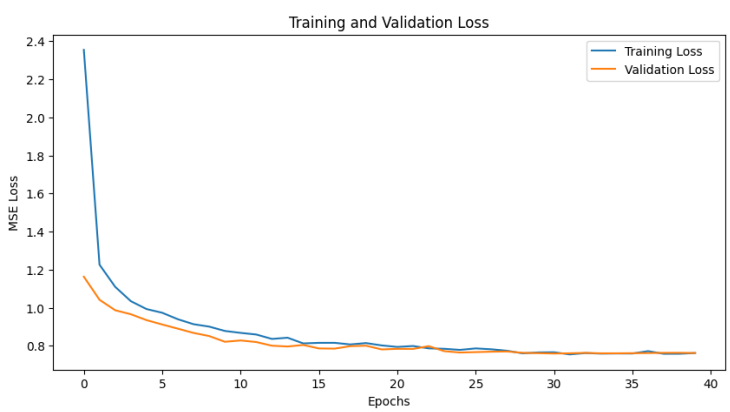


Chart 21: Training and Validation Loss (FNN)

$$MSE = 0.7245$$

This is slightly higher than that of XGBoost

Prediction Fit The prediction looks similiar to XGBoost, but seems slights more volatile and continuous:

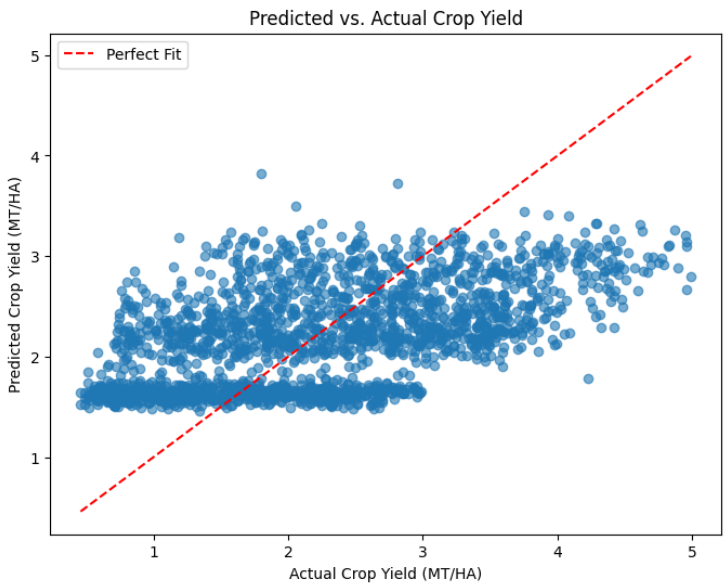


Chart 22: Actual vs. Predicted Crop Yield (FNN)

Feature Importance We use weights of the first layer to see feature importance of FNN, as seen below, although temperature remains significant, the other features plays more roles than in XGBoost.

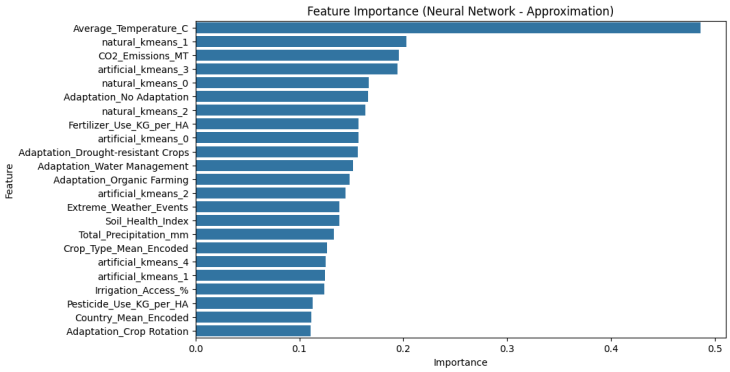


Chart 23: Feature Importance in FNN (Approximation)

Note: in ALL 3 models, a few of our engineered cluster features excels most other raw features, confirming the importance of our cluster-feature construction step.

5. Model Blending

Since we see that there is a difference in the weight of features applied by XGBoost and FNN, we wonder whether blending the model can result in more comprehensive, and hence lower variance results. We use the weighted blending approach. The blended predictions were calculated as:

$$BlendedPrediction = w_1 \cdot XGBoostPrediction + w_2 \cdot FNNPrediction$$

where $w_1 + w_2 = 1$. We perform a grid search of steps size 0.1 from (0,1) t0 (1,0)

Through analysis, the prediction MSE simply decreasingly converges to the MSE of XGBoost as XGBoost's weight approaches 1. This reveals blending does really works since XGBoost significantly outperforms FNN, so our blending fails to outperform XGBoost.

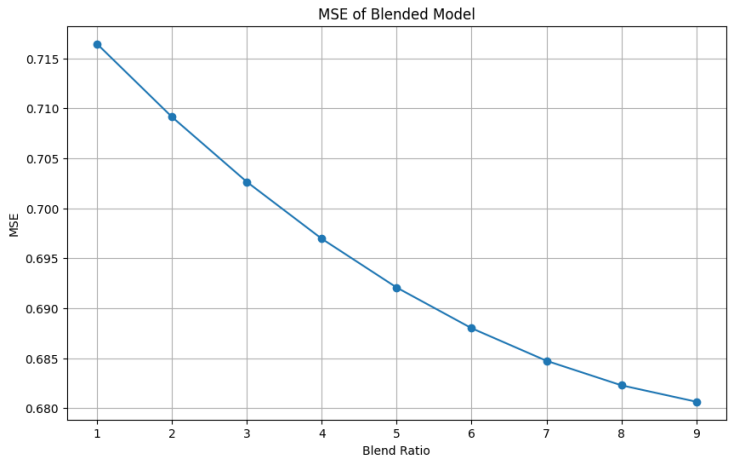


Chart 24: MSE Across Different Regions (XGBoost)

6. Regional-Specific Predictions

Since we are focusing on a global predictor, this might include unavoidable errors considering the variances among countries. Using the best solution we found above, XGBoosting with optimized hyperparamters, we perform prediction on each countries.

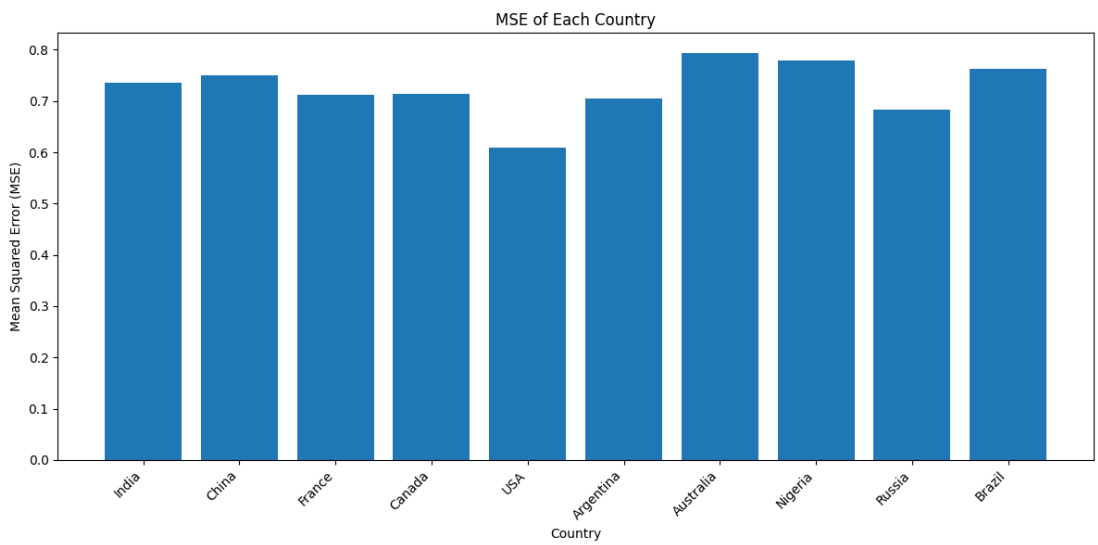


Chart 25: MSE Across Different Regions (XGBoost)

As we can see, the accuracy of countries seem to vary a lot, and country

specific-predictor does not outperforms the global one, the average MSE is even higher except for the US. This may be caused by the reduction of the number of data.

Conclusion

This project examined the interplay between natural and artificial factors and their impact on global agricultural yields. Using exploratory data analysis (EDA), clustering, and predictive modeling, we provided insights into the complex relationships influencing crop productivity.

Key Findings from Exploratory Data Analysis

- The EDA performs many basic analysis such as distribution of different types of crops, different countries, time series analysis.
- Correlation analysis showed weak pairwise correlations between individual features (e.g., temperature, precipitation, irrigation) and crop yields, suggesting that yields are influenced by non-linear combinations of multiple factors; Time series analysis suggests no serial correlations of yield with its past values.
- Geographical analysis highlighted region-specific trends, such as the higher rice yields in Northwestern India and Eastern Canada. These findings pointed to localized factors such as climate, soil health, and artificial inputs as critical determinants of productivity.

Insights from Clustering and Predictive Modeling

- Clustering analysis, using K-Means and Spectral Clustering, effectively grouped data based on both natural and artificial features. Clusters provided insights into common combinations of conditions and were leveraged as features in predictive models. K-Means on natural conditions proved particularly useful, creating clusters with the most significant yield differences.
- Predictive modeling highlighted XGBoost as the best-performing method, achieving an MSE of 0.68. This performance underscores the value of non-linear models for capturing complex interactions between features.
- Feature importance analysis identified temperature, CO_2 emissions, and precipitation as key drivers of yield variability, while artificial factors like irrigation access and fertilizer use showed regionally significant impacts.
- Blending models did not outperform XGBoost alone, confirming its robustness in this context, and regional specific predictor does not seem to outperform the general global predictor.

7 Advantages and Limitations

Advantages

- Visual tools such as heatmaps, time series, and bar charts improve interpretability and make the analysis accessible to stakeholders.
- Clustering methods like K-means and spectral clustering group similar environmental and artificial factors, enabling the creation of new predictive features and identifying combined influences on agricultural productivity.
- Comparing models, including Linear Regression, Neural Networks, and XGBoost, provides a comprehensive understanding of climate and artificial impacts, supporting actionable predictions to address food shortages.
- Feature importance analysis reveals localized threats to agricultural systems, guiding strategies such as improved irrigation, sustainable fertilizer use, and soil health restoration to enhance resilience and long-term yields, especially in resource-constrained regions.

Limitations

- The best predictor achieved a Mean Squared Error (MSE) of approximately 0.7, compared to the standard deviation of around 1, indicating that while the model explains some variability, a large portion remains unexplained.
- Attempts to blend models and create regional-specific predictors were relatively basic, suggesting the need for more in-depth approaches and analysis.
- The dataset contains too many combinations of regions and crops, with each specific combination having fewer than hundreds of observations, making it difficult to focus on targeted regions and crops.
- For better local decision-making, future efforts should focus on collecting more localized data for specific crops, which may reduce errors in general predictors and improve accuracy.
- Translating these results into actionable policies for farmers and governments requires additional contextual understanding and collaboration and needs further research and effort.

8 Group Contributions

Maya: Maya helped research the topic and pick the dataset for the project. She wrote the Introduction, Project Goals, as well as the entire Section 4.1 and references of the report. She wrote the corresponding code for the geographical analysis and wrote up summaries and comments in the code to help connect and

make the code more cohesive. She made the slides for the geographical section of the presentation and throughout the project, helped delegate, and completed tasks in a timely manner.

Jiaxuan: Jiaxuan works with analysis parts with respect to time series, performs feature engineering with standardization, encoding, and clustering. He implemented clustering and prediction part of the data. He is also in charge of reporting the prediction part in the presentation and the final report.

Fangyan: She contributed to the research on the topic, wrote the description of the dataset, and authored Sections 3 and 5 of the report. She created the feature correlation matrix, developed the data normalization code, and worked on early versions of clustering, linear regression, and some neural network models. She also completed the slides for the Introduction and Initial Exploration of the Dataset and Features.

9 References

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